5 **(a)** **(i)** Distinguish between electrical resistance and resistivity.

.....

.....

..... [1]

(ii) A metal wire XY of resistance 2.0 Ω has a diameter of 1.0 mm and a resistivity of 1.5×10^{-6} Ω m.

Calculate the length of the wire.

length = ... m [1]

(b) A battery of e.m.f. 6.0 V with negligible internal resistance is connected to the metal wire XY and a light bulb of resistance 4.0 Ω as shown in Fig 5.1. The length of the connecting wire joining the negative terminal of the battery to the lamp is 0.20 m.

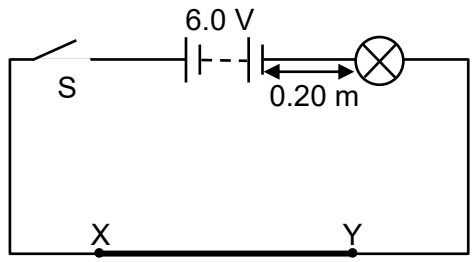

Fig. 5.1

(i) Switch S is closed. Calculate the current in the circuit.

current = ... A [1]

(ii) The connecting wires of diameter 1.0 mm are made of copper.

Given that the density and molar mass of copper are 8.96×10^3 kg m^{-3} and 64.0 g respectively, calculate the average drift velocity in the copper wires.

Assume that the number of conduction electrons is equal to the number of copper atoms in the wire.

drift velocity = ... m s^{-1} [3]

(iii) Calculate the time it would take for an electron to move from the negative terminal of the battery to the light bulb.

time = ... s [1]

(iv) The light bulb lights up in a time much lesser than the time calculated in **(b)(iii)**. Explain this observation.

.....

.....

.....

..... [2]

(c) The light bulb is removed from the circuit in Fig. 5.1.

A cell with e.m.f. 3.0 V and internal resistance 0.50 Ω and a galvanometer are now connected to the circuit as shown in Fig. 5.2.

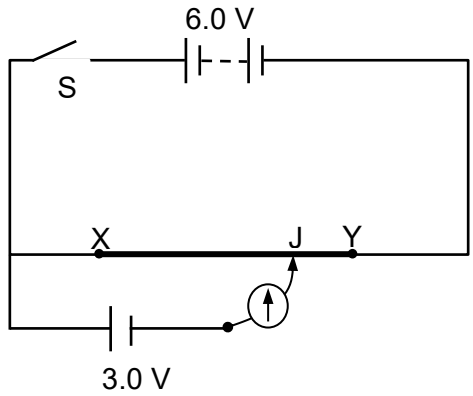

Fig. 5.2

(i) Calculate the length XJ when the galvanometer reads zero.

length XJ = ... m [1]

(ii) A 1.0 Ω resistor is now connected across the 3.0 V cell.

Calculate the new length XJ when the galvanometer reads zero.

length XJ = ... m [2]